李盼林（直博申请）

高维统计学习 | 数据建模

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中国工业与应用数学学会会员 · 山东省青年摄影家协会会员 · 青岛市青年摄影家协会会员

教育背景

新疆农业大学（数理学院）　数学与应用数学 · 本科 (2.5%)　导师：黄华教授　2023 – 2027

- 荣誉奖项：共青团中央“优秀志愿者”（9 次）及“优秀团队”、“少年儿童安全守护计划”优秀团队（负责人）；新疆农业大学“三好学生”“创新创业之星”（连续两年）、“科创先锋奖学金”。
- 核心成果：已发表 11 篇学术论文（含 8 篇一作），另有 3 篇一区 SCI 一作在投；主持或参与 16 项大学生创新项目（国家级 3 项、自治区级 6 项），项目获华为研究院认可；持有 4 项实用新型专利、4 项软件著作权；获 5 项国家级、7 项自治区级竞赛奖项；志愿事迹被《中国日报》《人民日报》等多家媒体报道。

科研项目

全国统计科学研究项目 · 重大项目 · 申请中　棉花产量与气候风险的多模态高维统计学习方法研究　2026 – 今

- 主笔撰写国家级重大项目申报书与 7000 字综述；设计” 异构-多模态张量稀疏融合”、” 时空双重去偏 Lasso + 广义随机森林” 因果识别框架与多流 INSPECT + PCMCI 棉田预警方案，申报书已提交评审。

新疆自科基金面上项目 · 参与　高光谱成像数据空谱特征融合的函数型统计建模方法及农产品智能检测应用　2026 – 今

- 整合多品种 HSI/NIR 数据，完成 SNV/MSC 校正与多基底函数型表征；围绕函数型对比稀疏多任务、机理嵌入采后建模、农残光谱筛查形成三篇一作论文初稿（导师审核中）。

国家级大创 · 在研 · 参与　基于机器学习与迁移学习的苹果溯源模型研究　2025 – 今

- 主笔申报书，牵头采集新疆/山东/甘肃跨年苹果 HSI 与图像数据；构建 9 域基准与” 产地 × 年份” 防泄漏协议，完成 PLS-DA/SVM/1D-CNN/Transformer 对比；一作投稿 CEA（SCI 一区 Top, IF: 8.9）。

其他项目：能源经济小样本时序预测（国家级，结项，负责人）· 文脉智渊 Hub（自治区级，优秀结项，负责人）· 神机棉算数字棉田（自治区级，在研，负责人）· 瞳曦智行多模态视障导航（自治区级，在研）· 函数型对比稀疏多任务高光谱检测（自治区级，在研）· 函数型数据 + 图像融合苹果 SSC 预测（自治区级，优秀结项）· 高维非线性 PDE 波解（校级，优秀结项，负责人）· 智瞳明杖智能盲杖（校级，优秀结项）等。

科研成果

- Li P, Li Y C, Huang H, et al. *Functional Multi-Task Sparse Learning with Spectral-Spatial Information Fusion for Hyperspectral Food Quality Detection*. Information Fusion.（SCI 一区 Top, IF: 15.5, 在投）
- Li P, Li L J, Liu Y, et al. *Robust cross-origin near-infrared prediction of soluble solids content in apples via a Beer–Lambert physics-informed prior*. Food Chemistry.（SCI 一区 Top, IF: 9.8, 在投）
- Li P, Zhang Q L, Jiao Q T, et al. *Cross-year declared-origin discrimination of commercial apples by near-infrared spectroscopy: a nine-domain benchmark, a leakage-controlled evaluation protocol, and a baseline study*. Computers and Electronics in Agriculture.（SCI 一区 Top, IF: 8.9, 在投）
- 其他论文：NIPT 检测时点优化（IJP HMR, 2026）；KdV 方程周期解（应用数学进展, 2025）；稳健贝叶斯岭回归预测 NIPT 时点（AJCIS, 2025）；多模型回归判别胎儿染色体异常（AJMS, 2025）；PCA 高维数据优化（ICCECT 2025）；灰色模型改进（TCAM, 2024，共同一作）；扩展 (3+1) 维浅水波方程孤波解与相互作用解（长春师范大学学报, 2026）；ESP32 多传感器水质监测系统（物联网技术, 2026）；图像颜色特征与 Vis-NIR 光谱融合预测苹果 SSC（光谱学与光谱分析, 2025）等。
- 知识产权：实用新型专利 4 项（导盲杖节连接组件、便携式超声波探测装置、喷洒农药无人机、可折叠防撞喷药杆无人机）；软件著作权 4 项。

竞赛奖项

- 正大杯第 16 届全国大学生市场调查与分析大赛 · 本科生组三等奖
- 第 11 届全国大学生统计建模大赛 · 本科生组三等奖
- 正大杯第 15 届全国大学生市场调查与分析大赛 · 本科生组三等奖
- 中国国际大学生创新大赛（2024）· 铜奖
- 数据蜂杯全国大学生访谈调查大赛 · 三等奖
- 第 14 届中国创新创业大赛新疆赛区 · 成长组三等奖
- 中国国际大学生创新大赛（2025）新疆赛区 · 铜奖
- 第 11 届全国大学生统计建模大赛新疆赛区 · 一等奖
- 正大杯第 15 届全国大学生市场调查与分析大赛新疆赛区 · 一等奖
- 2024 年新疆赛区信息素养大赛 · 优秀奖
- 2024 年全国大学生数学建模竞赛新疆赛区 · 二等奖
- 第 10 届全国大学生统计建模大赛新疆赛区 · 二等奖

学生经历

数理学院数学 **2302** 班 班长　2023.09 – 2027.07

新疆农业大学第十九期青年马克思主义者培训班 班长　2025.05 – 2026.05

新疆农业大学数理学院志愿者协会 负责人　2024.09 – 2025.11

- 任班长期间组织学风建设与体育竞赛，带领班级获 2024–2025 学年新疆农业大学优秀班集体与体育先进班级。
- 发起并组建” 未来之盾” 大学生志愿团队（20 人），加入中国青年志愿者协会；参与 11 项共青团中央与中国青年志愿者协会主办的支教、乡村振兴等公益项目，足迹覆盖山东、河南、新疆等地的市县；事迹被《中国日报》《人民日报》等多家媒体报道；团队获国家级优秀青年志愿者团队称号，之后受邀请加入数理学院志愿者协会。

个人技能

- 外语能力：托福 110/120；以第一作者发表多篇英文 SCI/IEEE 学术论文，具备独立的英文学术写作能力。
- 专业能力：高维统计推断（去偏 Lasso、双重机器学习、广义随机森林）、函数型数据分析（函数型 PCA、函数对函数回归）、张量分解与多模态融合；熟练使用 Python（PyTorch / scikit-learn），能独立完成” 理论推导 → 算法实现 → 实证验证” 端到端建模。
- 工具链：LaTeX（本简历即由 LaTeX 撰写）、Git 版本控制、Lightroom / Final Cut Pro 影像处理。

自我认知

- 摄影、自由、大自然；喜欢接触新东西，愿意研究，动手能力强，执行力强，信息整合力强，探索欲旺盛。
- 本科借 16 个项目与 11 篇论文走通数学 → 统计 → 机器学习 → 真实场景的闭环，清楚短板在理论深度；直博阶段希望从应用驱动走向更理论的方法研究，同时保持对落地应用的敏感度。

Panlin Li (Direct Ph.D. Application)

High-Dimensional Statistical Learning | Data Modeling

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https://2004lryan.github.io/

Member, China Society for Industrial and Applied Mathematics · Member, Shandong Young Photographers' Association · Member, Qingdao Young Photographers' Association

Education

Xinjiang Agricultural University (School of Mathematics & Physics) | Mathematics & Applied Mathematics · B.S. (Top 2.5%) | Advisor: Prof. Hua Huang | Sep. 2023 – Jul. 2027

- Honors: Outstanding Volunteer (9×) & Outstanding Team — CYL Central Committee; Hematopoietic Stem Cell Donation Honor Certificate; Outstanding Student, Innovation & Entrepreneurship Star (2 consecutive years), Science & Innovation Pioneer Scholarship — XJAU.
- Key outcomes: led/participated in 16 student innovation projects (3 national, 6 provincial), project recognized by Huawei Research Institute; first-authored 8 papers (plus 3 Q1 SCI under review); 4 utility-model patents, 4 software copyrights; 5 national & 7 provincial competition awards; volunteer work covered by *China Daily*, *People's Daily*, and other media outlets.

Research & Projects

National Statistical Science Research Project · Major Program · Under Application — Multimodal High-Dimensional Statistical Learning for Cotton Yield and Climate Risk — 2026 – Present

- Responsibilities: lead-authored the National Major Program proposal and a 7,000-word survey under advisor guidance; proposed a heterogeneous-multimodal tensor-sparse fusion model jointly modeling meteorological / soil / remote-sensing / phenological signals; designed a spatio-temporal doubly-debiased Lasso + generalized random forest causal-identification framework; planned a multi-stream INSPECT + PCMCI field-level early warning system.
- Key outcomes: proposal submitted and under review.

Xinjiang Natural Science Foundation · General Program · Participant — Functional Statistical Modeling via Spatial-Spectral Feature Fusion of Hyperspectral Imaging — 2026 – Present

- Responsibilities: curated multi-cultivar HSI/NIR public datasets; performed SNV/MSC correction and multi-basis functional representation; developed three algorithmic directions — functional contrastive sparse multi-task, mechanism-embedded post-harvest modeling, and residue spectral screening; lead-authored three corresponding papers.
- Key outcomes: three first-author drafts completed (under advisor review).

National Innovation Project · Ongoing · Participant — Apple Origin Traceability via Machine Learning & Transfer Learning — 2025 – Present

- Responsibilities: lead-authored the proposal; led cross-year apple HSI and image data collection across Xinjiang/Shandong/Gansu; built a 9-domain benchmark with an origin × year double-layered leakage-controlled evaluation protocol; ran PLS-DA / SVM / 1D-CNN / Transformer baselines and cross-year generalization study; led paper writing and submission.
- Key outcomes: first-author submission to *Computers and Electronics in Agriculture* (SCI Q1 Top; IF: 8.9).

Other Projects: Small-Sample Time-Series Forecasting (National, Completed, PI) · Wenmai Zhiyuan Hub (Regional, Excellent, PI) · Shenji Miansuan — Digital Cotton Field (Regional, Ongoing, PI) · Tongxi Zhixing — Multimodal Navigation (Regional, Ongoing) · Functional Contrastive Sparse Multi-Task HSI Detection (Regional, Ongoing) · Apple SSC Prediction via FDA+Image Fusion (Regional, Excellent) · Nonlinear Wave Solutions of High-Dimensional PDEs (University, Excellent, PI) · Zhitong Smart Cane (University, Excellent) · Tianshan Pastoral Song (University, Excellent) · Non-Traveling-Wave Solutions of Nonlinear PDEs (University, Pass) · "Cure on Arrival" — AI Pest Diagnosis (University, Pass) · Online Rapid Detection via Spectral Transfer (University, Ongoing, PI) · Fruit & Vegetable Storage Quality Evolution (University, Ongoing)

Publications

- Li, Panlin; Li, Yuchang; Huang, Hua; et al. *Functional Multi-Task Sparse Learning with Spectral-Spatial Information Fusion for Hyperspectral Food Quality Detection*. Information Fusion. (SCI Q1 Top; IF: 15.5; Under Review)
- Li, Panlin; Li, Longjie; Liu, Ya; et al. *Robust cross-origin near-infrared prediction of soluble solids content in apples via a Beer–Lambert physics-informed prior*. Food Chemistry. (SCI Q1 Top; IF: 9.8; Under Review)
- Li, Panlin; Zhang, Qingli; Jiao, Qitai; et al. *Cross-year declared-origin discrimination of commercial apples by near-infrared spectroscopy: a nine-domain benchmark, a leakage-controlled evaluation protocol, and a baseline study*. Computers and Electronics in Agriculture. (SCI Q1 Top; IF: 8.9; Under Review)
- Li P, Zhang N. *Optimization of NIPT Testing Timing and Fetal Anomaly Determination Models*. International Journal of Public Health and Medical Research, 2026, 6(2): 1–8.
- Li, Panlin; Bao, Fangchen; Zhang, Jingyi; et al. *Periodic Solutions to a New Class of KdV Equations*. Advances in Applied Mathematics, 2025, 14(5): 23–28.
- Li P, Zhang N. *Prediction of optimal NIPT timing and analysis of influencing factors of fetal Y chromosome concentration based on robust Bayesian ridge regression*. Academic Journal of Computing & Information Science, 2025, 8(10): 99–107.
- Li P, Zhang N. *Study on the determination of fetal chromosomal abnormalities based on multi-model regression*. Academic Journal of Mathematical Sciences, 2025, 6(2): 121–129.
- Li P, Zhang N. *Study on the effect of n-hexane insolubles on pyrolysis product yields based on statistical regression analysis*. Highlights in Science, Engineering and Technology, 2024, 116: 270–276.
- Li P, Zhang N. *Research on Optimizing High-dimensional Data Model Based on Principal Component Analysis*. In: 2025 IEEE 3rd International Conference on Control, Electronics and Computer Technology (ICCECT), Jilin, China, 2025, pp. 998–1004.

- Li, Panlin; Li, Jie; Alamusijiang Kadir; et al. *Cultural Heritage and Rural Revitalization: Exploration and Reflections on China's Practice*. Smart Agriculture Guide, 2024, 4(21): 185–188.
- Rao W J, Li P L. *An improved kernel regularized nonhomogeneous grey model based on conformable fractional accumulation*. Transactions on Computational and Applied Mathematics, 2024, 4: 137–147. (Co-First Author)
- Zhang, Ning; Li, Panlin; Bao, Fangchen; Liu, Xiaoyu. *Solitary Wave Solutions and Interaction Solutions of an Extended (3+1)-Dimensional Shallow Water Wave Equation*. Journal of Changchun Normal University, 2026, 45(4): 6–11.
- Jiao, Qitai; Li, Panlin; Mei, Tao; et al. *ESP32-Based Multi-Sensor Near- & Far-Range Water Quality Monitoring System*. Internet of Things Technologies, 2026, 16(2): 9–12, 16.
- Huang, Hua; Wang, Mo; Liu, Ya; Li, Panlin; et al. *Prediction of Apple Soluble Solids Content Using a Functional Partial Linear Regression Model Fusing Image Color Features and Vis-NIR Spectra*. Spectroscopy and Spectral Analysis, 2025, 45(S1): 415–423.
- **Patents (4, granted):** Section-Connection Component for a White Cane (CN2024230029031) · Portable Ultrasonic Detection Device (CN2024230026103) · Pesticide Spraying Drone (CN20241226759) · Foldable Anti-Collision Spraying Boom for Drones (CN20241226785).
- **Software Copyrights (4):** Intelligent White Cane Vision Recognition System V1.0 · Smart Cane Data Collection & Monitoring System V1.0 · User Behavior Analysis and Recommendation System V1.0 · Big-Data-Driven Market Analysis and Forecasting System V1.0.

Competitions & Awards

- "Chia Tai Cup" 16th National College Students Market Survey & Analysis Contest · Undergraduate Third Prize *Team Lead* 2026.05
- 11th National Undergraduate Statistical Modeling Contest · Undergraduate Third Prize *Team Lead* 2025.10
- "Chia Tai Cup" 15th National College Students Market Survey & Analysis Contest · Undergraduate Third Prize *Team Lead* 2025.05
- China International College Students' Innovation Competition (2024) · Bronze Award *Participant* 2024.10
- DataBee Cup National College Student Interview Survey Contest · Third Prize *Participant* 2024.10
- 14th China Innovation & Entrepreneurship Competition Xinjiang Division (12th Xinjiang Competition) · Growth Group Third Prize *Team Lead* 2025.10
- China International College Students' Innovation Competition (2025) Xinjiang Division · Bronze Award *Team Lead* 2025.08
- 11th National Undergraduate Statistical Modeling Contest Xinjiang Division · Undergraduate First Prize *Team Lead* 2025.08
- "Chia Tai Cup" 15th National College Students Market Survey & Analysis Contest Xinjiang Division · Undergraduate First Prize *Team Lead* 2025.04
- 2024 Xinjiang Division Information Literacy Competition · Excellence Award *Team Lead* 2024.12
- 2024 National College Students Mathematical Modeling Contest Xinjiang Division · Undergraduate Second Prize *Team Lead* 2024.11
- 10th National Undergraduate Statistical Modeling Contest Xinjiang Division · Second Prize *Team Lead* 2024.07

Student Experience

Class Monitor, Mathematics 2302, School of Mathematics & Physics 2023.09 – 2027.07
Class Monitor, 19th Youth Marxist Training Program, Xinjiang Agricultural University 2025.05 – 2026.05
Head, Volunteer Association, School of Mathematics & Physics, XJAU 2024.09 – 2025.11

- Organized academic and sports activities; led the class to receive *Outstanding Class Collective* and *Sports Pioneer Class* honors (AY 2024–2025).
- Founded the "Shield of the Future" volunteer team (20 members); participated in 11 CYL Central Committee / CYVA public-welfare programs across Shandong, Henan, and Xinjiang; work reported by *China Daily*, *People's Daily*, and other media outlets.

Skills

- **Language:** TOEFL 110/120; first-authored multiple English SCI/IEEE papers.
- **Professional:** high-dimensional statistical inference (debiased Lasso, double ML, generalized random forests), functional data analysis (FPCA, function-on-function regression), tensor decomposition & multimodal fusion; Python (PyTorch / scikit-learn), end-to-end modeling.
- **Tools:** LaTeX, Git, Lightroom / Final Cut Pro.

Self-Assessment

- Photography, freedom, the great outdoors; badminton (enthusiastic but mediocre); can hold a drink (never been drunk). Love diving into new things, eager to figure stuff out — strong hands-on, execution, and information-synthesis skills, driven by a deep exploratory urge.
- Through 16 projects and 11 papers during undergrad, I have walked the full loop from math → statistics → ML → real-world deployment — which has made me clearly aware that my bottleneck lies in theoretical depth. In the Ph.D. stage, I aim to move toward more theoretical directions while staying grounded in real-world applications, finding balance between methodological rigor and practical impact.